



ELISA

Enabling **Linux** in
Safety Applications

ELISA Workshop London, UK

June 9-11, 2026
Co-hosted with Canonical





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Safety Applications

Certifying Linux for Safety-Critical Systems: A Cybersecurity Compliance Case Study

Mikel Azkarate-askatsua

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Agenda



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- Dependability Journey
- Security Foundations and Compliance Strategy
- ISO 21434 (Automotive)
- IEC 62443-4-1 (Industrial)
- What about a Linux Distro?
- Success Stories
- Next Steps

Dependability Journey



Dependability Journey - Canonical's Mission

To amplify the impact of open source by making it secure, reliable and available to all for use at scale.

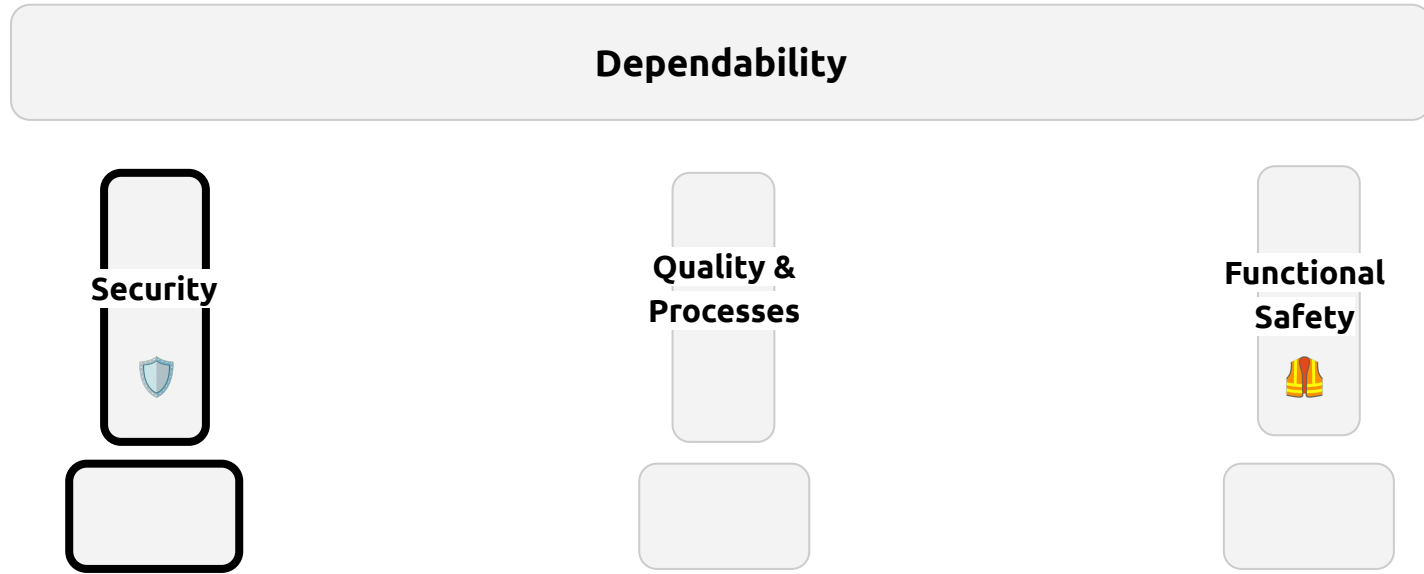
Dependability Journey - Canonical's Mission

*To amplify the impact of open source by making it **secure**, **reliable** and **available** to all for use at scale.*

Dependability Journey - Pillars



Dependability Journey - Pillars

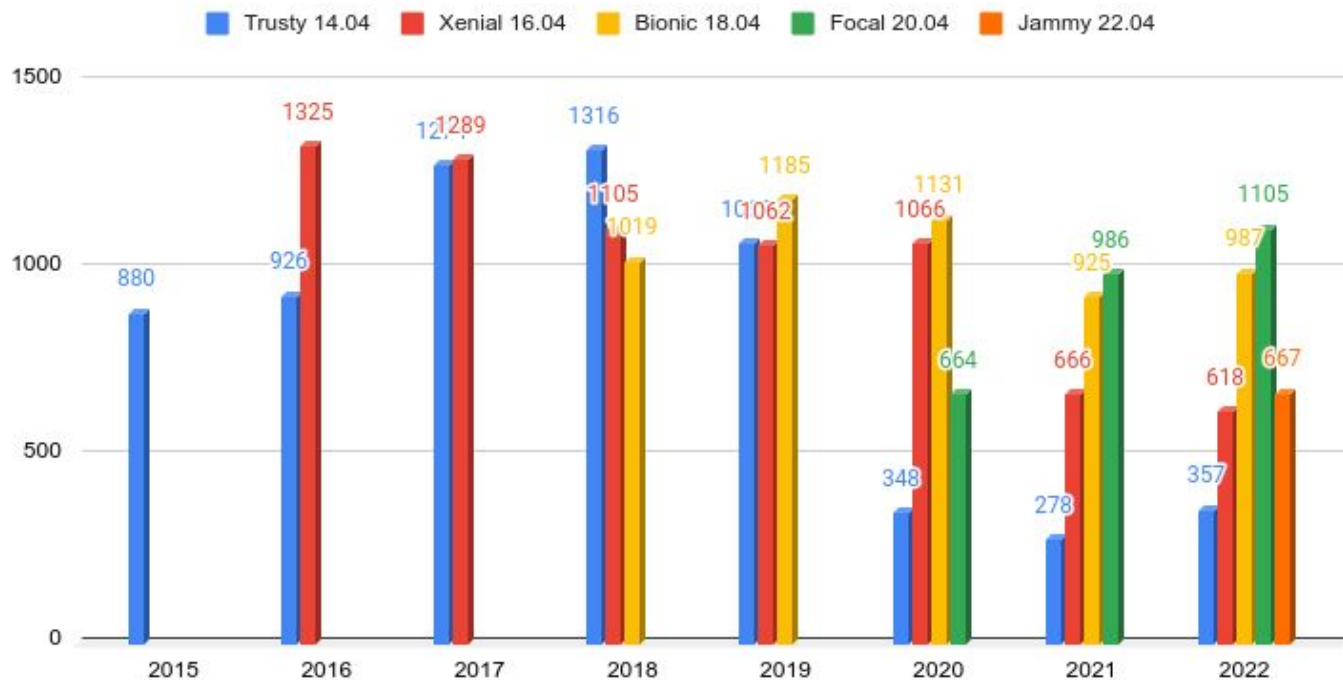


Security Foundations and Compliance Strategy



Security Foundations

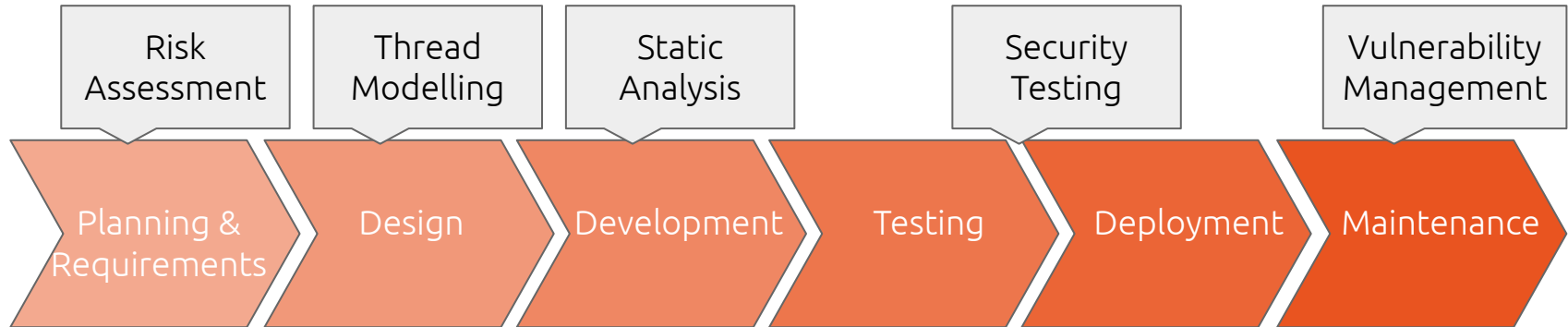
CVE fixes vs Ubuntu LTS



2015 ~ 2022 CVE fixes per Ubuntu LTS

Work in Progress - License: CC-BY-4.0

Security Foundations - Canonical's SSDLC



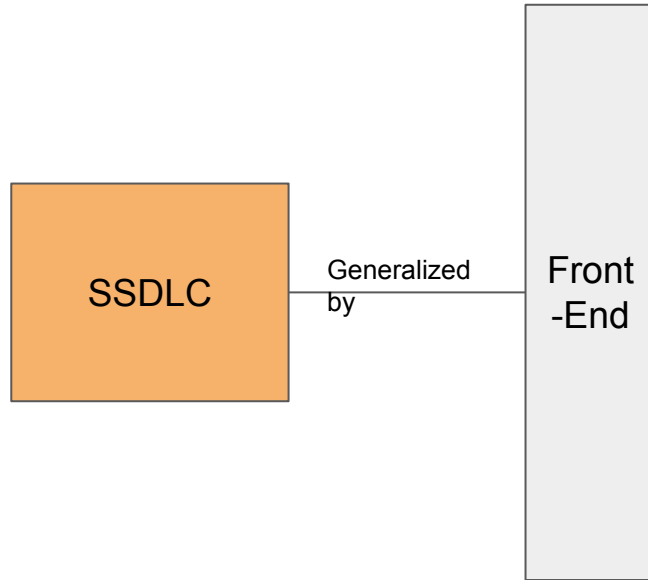
Compliance Strategy

- Targeting ISO 21434 (Automotive) and IEC 62443-4-1 (Industrial) Cybersecurity Management Systems (CSMSes)
- Do “the right thing” not just check the boxes
- Strategy?
 - One SSDLC per targeted standard?
 - Shall all Canonical products be aware of the Cybersecurity compliance framework?
 - What about the gaps?

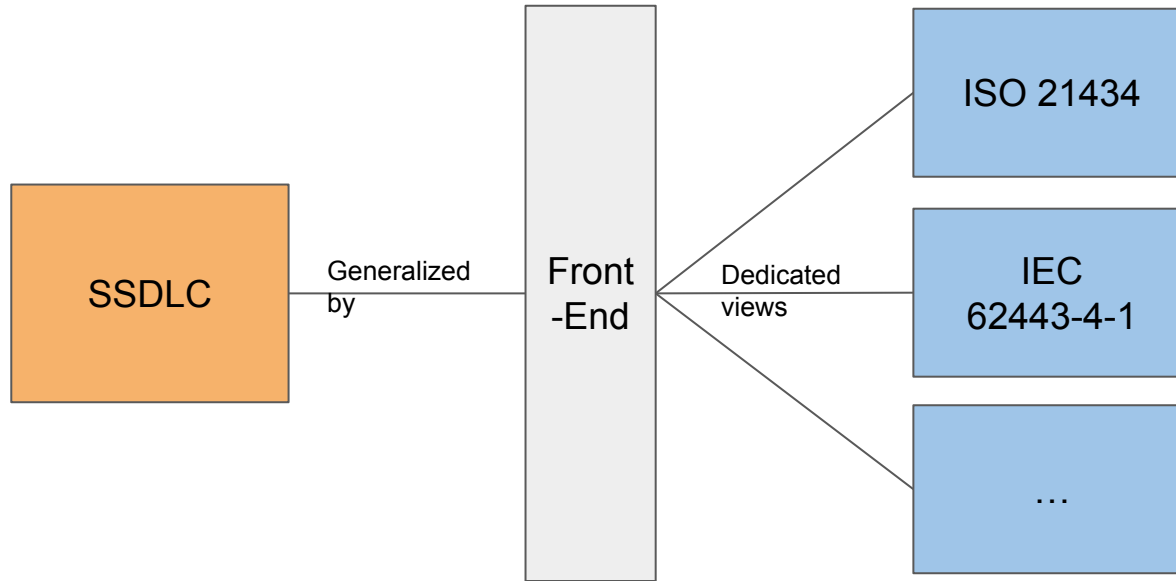
Compliance Strategy



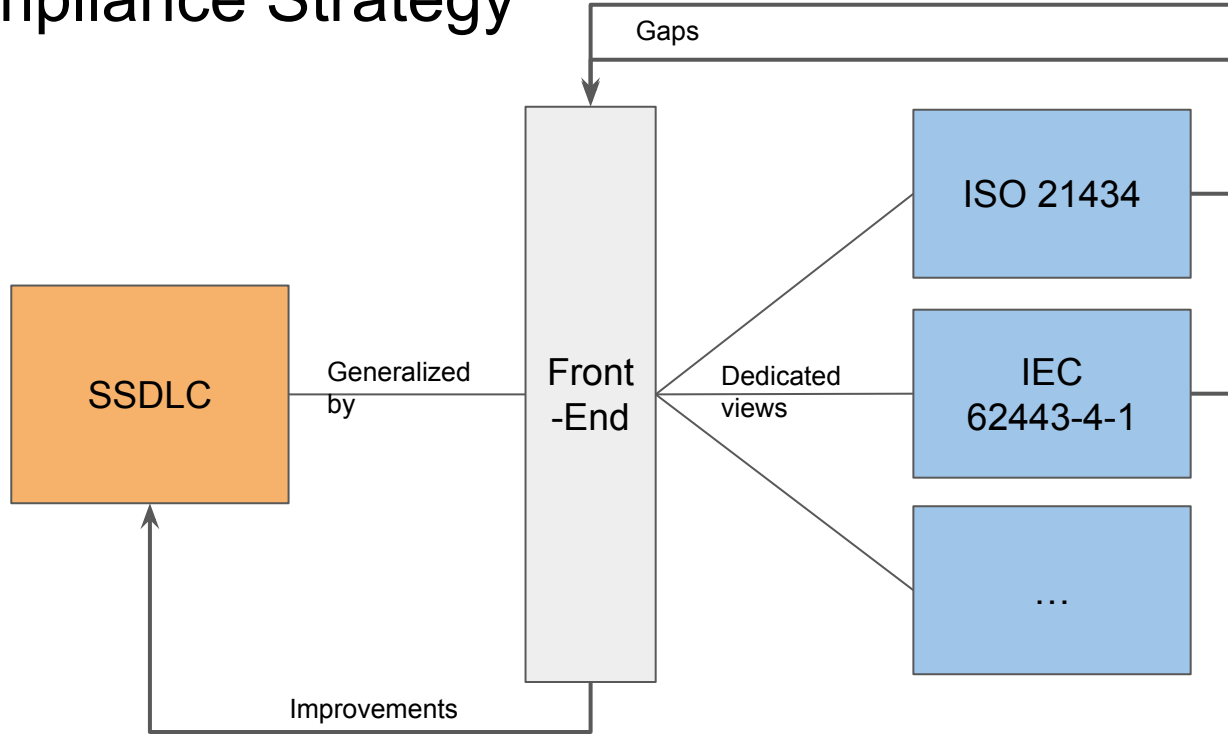
Compliance Strategy



Compliance Strategy



Compliance Strategy



ISO 21434 (Automotive)



ISO 21434 - Overview

- 5. Organizational cybersecurity management
- 6. Project dependent cybersecurity management
- 7. Distributed cybersecurity
- 8. Continual cybersecurity activities
- 9. to 14. Lifecycle phases
- 15. Threat analysis and risk assessment methods

ISO 21434 - Overview

- 5. Organizational cybersecurity management
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SSDLC

ISO 21434 - Overview

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SSDLC

ISO 21434 - Overview

5. Organizational cybersecurity management

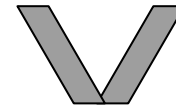
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SSDLC

IEC 62443-4-1 (Industrial)



IEC 62443-4-1 - Overview

Practice 1 - Security management

Practice 2 - Specification of security requirements

Practice 3 - Security by design

Practice 4 - Security implementation

Practice 5 - Security verification and validation testing

Practice 6 - Management of security related issues

Practice 7 - Security update management

Practice 8 - Security Guidelines

IEC 62443-4-1 - Overview

Practice 1 - Security management

SSDLC

Practice 2 - Specification of security requirements

Practice 3 - Security by design

Practice 4 - Security implementation

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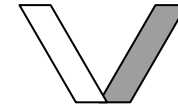
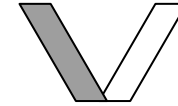
Practice 5 - Security verification and validation testing

Practice 6 - Management of security related issues

Practice 7 - Security update management

Practice 8 - Security Guidelines

SSDLC



What about the Linux Distro?



Ubuntu as a showcase

- Ubuntu Archives:
 - **main** are packages that can be considered our standard offering, these are the packages that Canonical provides security updates for and the commercial technical support.
 - **universe** are packages built from a range of public sources. Canonical provides regular security updates for software in universe only under Ubuntu Pro subscription.
 - **multiverse** ...
 - **restricted** ...

Ubuntu - Main Inclusion Review

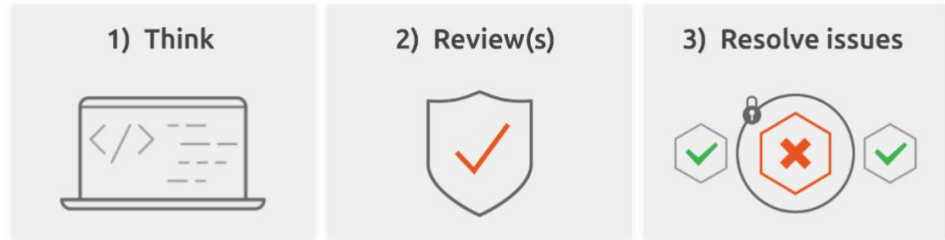
- “Off-the shelf component” - ISO 21434
- “Security requirement for externally provided components” - IEC 62443-4-1

Ubuntu - Main Inclusion Review

- “Off-the shelf component” - ISO 21434
- “Security requirement for externally provided components” - IEC 62443-4-1

E.g.: All new packages that are added to **main** must be added by following the **Main Inclusion Process (MIR)**. MIR is a largely manual process during which reviewers analyse and validate various aspects of the package, examples of criteria used in MIR:

- Security
- Maintainability
- Installability, buildability
- Autopkgtest coverage



Ubuntu - Pocket Management

Pocket refers to a way of segregating package versions into different groups for different purposes when preparing them for a release

- **Proposed** pocket is a staging area for changes to be included in updates or release. Updates for an upcoming release are staged and tested in proposed, and then incorporated in the release. Updates for an existing release go through a Stable Release Process (SRU), staged in proposed and then incorporated in updates.
- **Release** pocket is used during the development cycle for preparing the packages for a release. Once the development version is released, the release pocket is frozen and does not change
- **Updates** pocket includes critical bug fixes for packages in an already released version. Only the most critical bugs are fixed in already released Ubuntu versions.
- **Security** pocket includes updated packages with security-related fixes without any dependencies from updates. Security fixes are incorporated in the existing releases covered by our support commitments and the upcoming release.

Ubuntu - Vulnerability Management

- “Continual cybersecurity activities” - ISO 21434
- “Management of security related issues and Security update management” - IEC 62443-4-1

Ubuntu - Vulnerability Management

- “Continual cybersecurity activities” - ISO 21434
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E.g. Vulnerability Management process:

1. Learning about vulnerabilities
2. Triage
3. Coordination
4. Remediation
5. Disclosure & Communication



Success Stories



Canonical CSMS certified



Canonical

on 6 February 2025

Canonical achieves ISO/SAE 21434 certification, strengthening automotive cybersecurity standards



Canonical

on 18 September 2025

Canonical achieves IEC 62443-4-1 compliance in Industrial Automation and Control Systems

Share on:   

Ubuntu IEC 62443-4-2 (Product) Certified



Civil Infrastructure Linux - IEC 62443

- Some joint work on IEC 62443 vs. Linux
- Existing Gitlab site with resources:
 - <https://gitlab.com/cip-project/cip-documents>

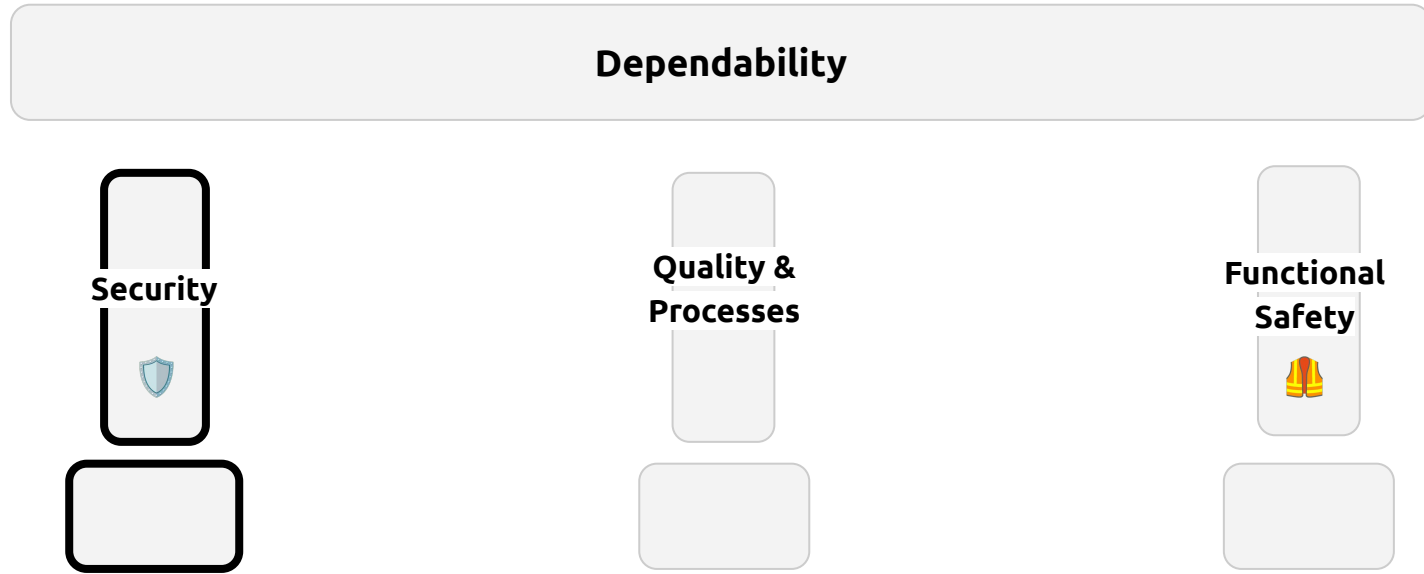


CIVIL
INFRASTRUCTURE
PLATFORM
INDUSTRIAL GRADE LINUX

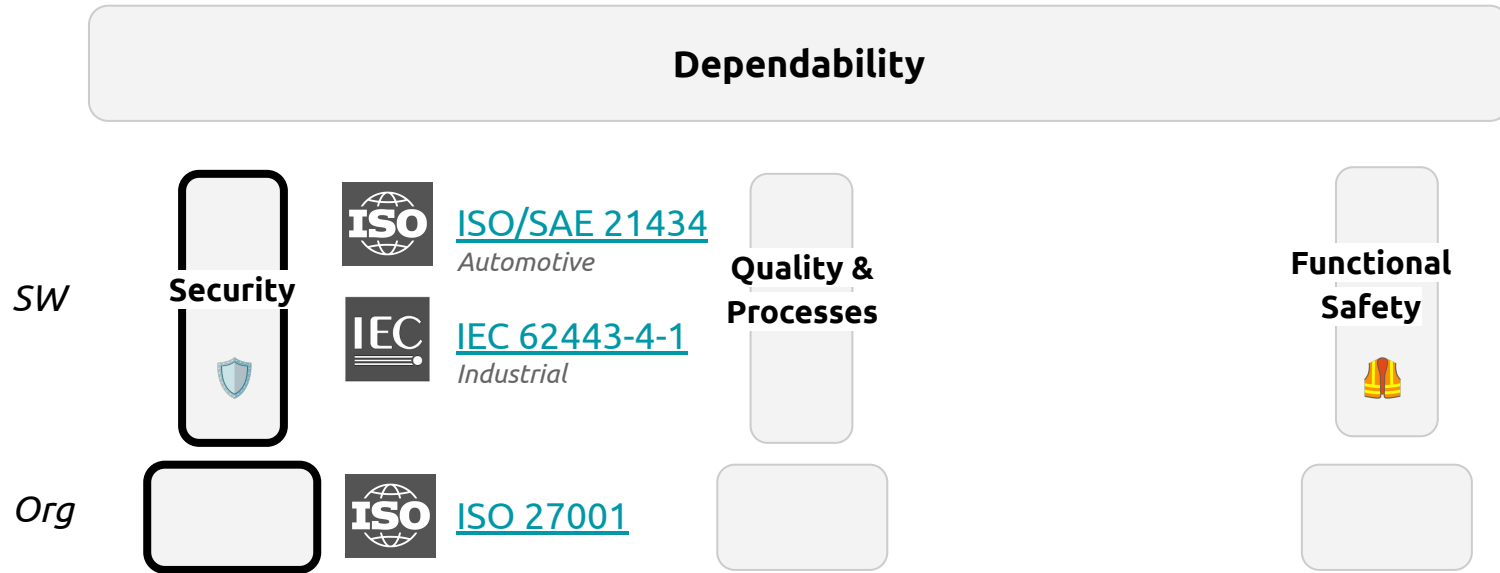
Next Steps



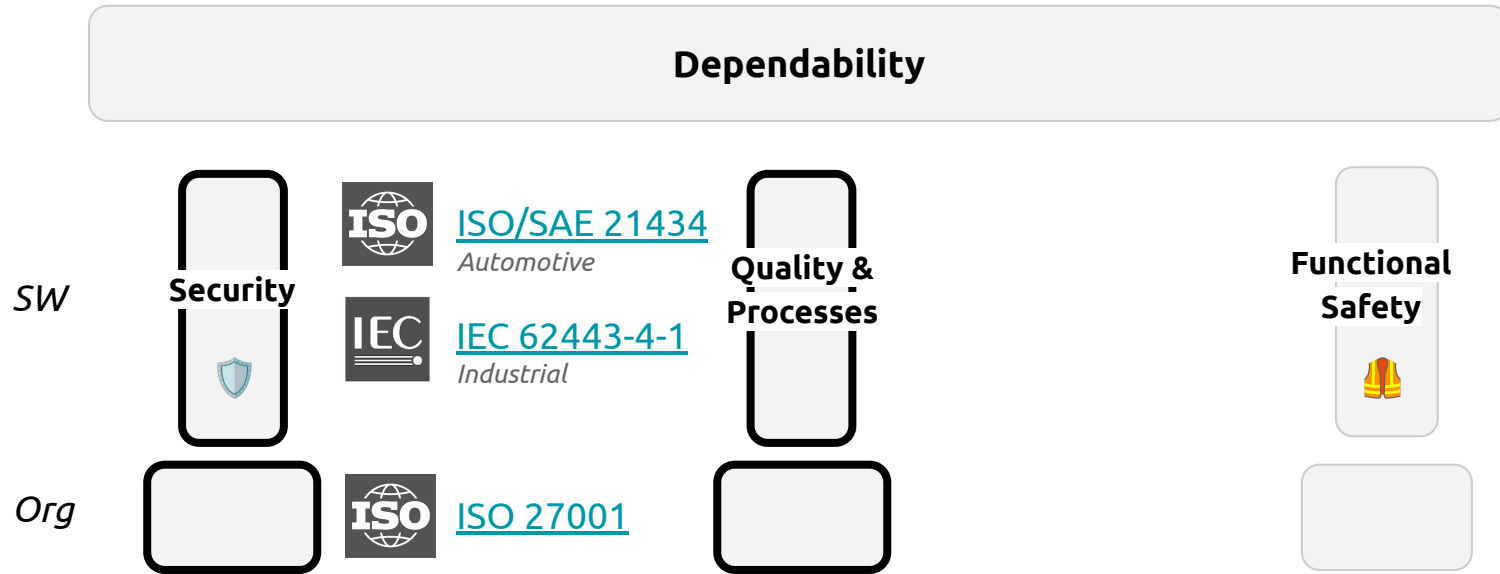
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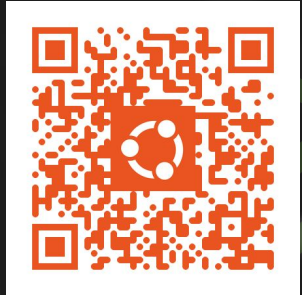


Dependability Journey - Pillars



Thank you! Questions?

We are hiring:



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